

ISE549 Management of Innovation and Technology



Knowledge, Learning & Innovation, Critical Success Factors for Innovation projects

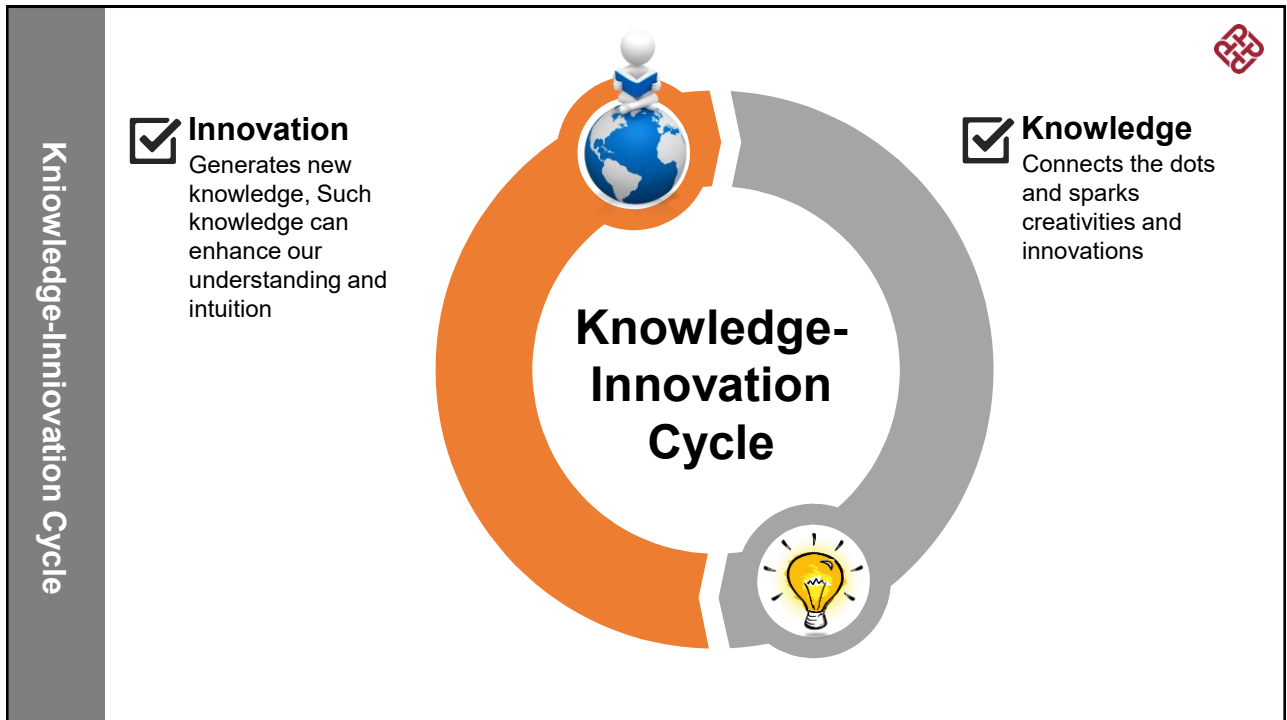
Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

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Economic Growth

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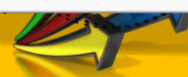
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Innovation

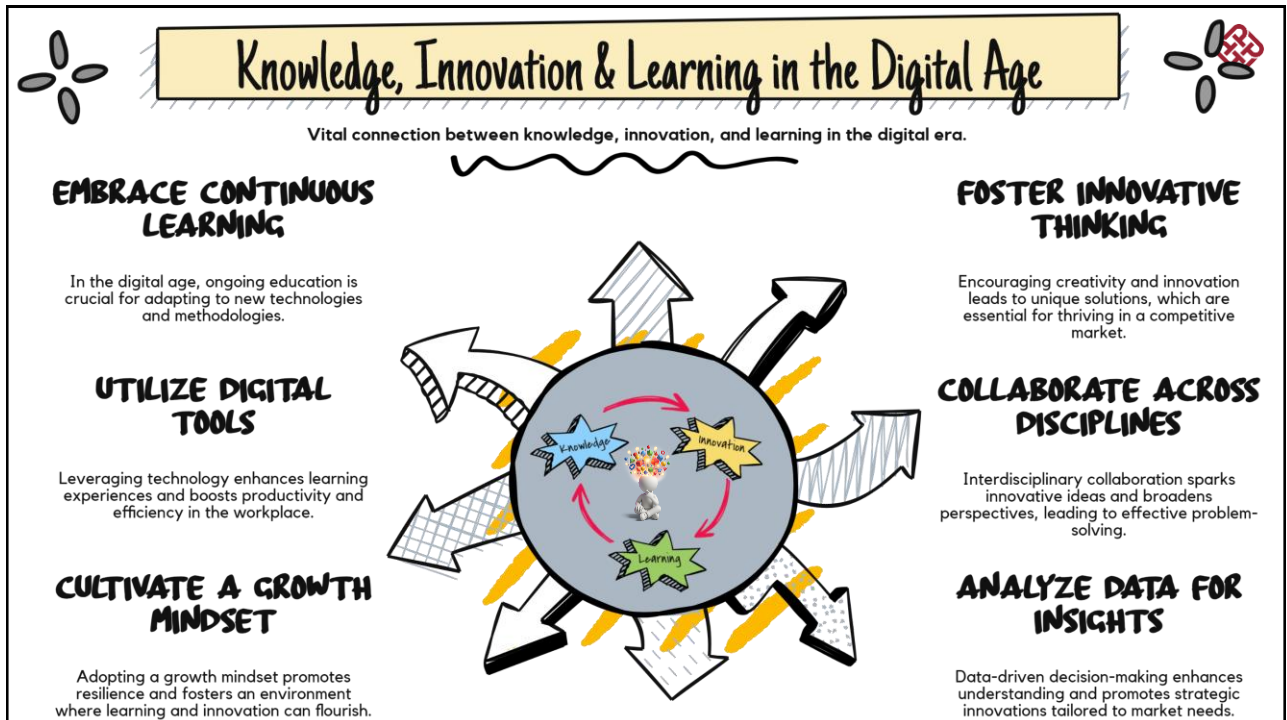
New Research on the Link Between Learning and Innovation

by Jean-François Harvey, Johnathan R. Cromwell, Kevin J. Johnson and Amy C. Edmondson

July 14, 2025

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KM efforts to spawn innovation & technological advancements

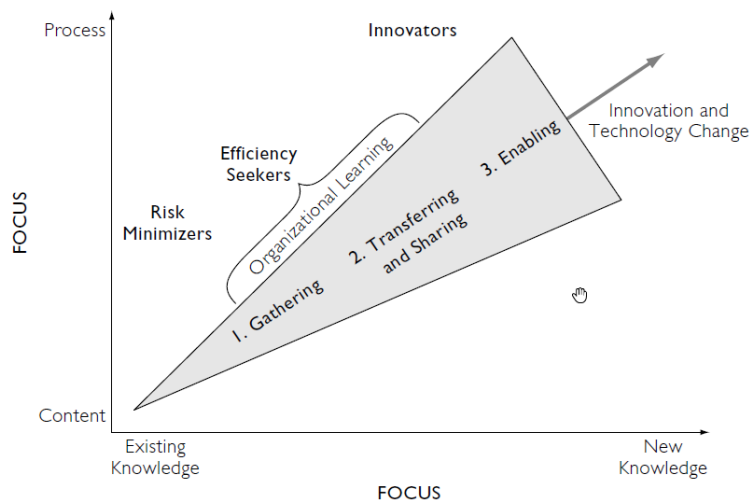


FIGURE 10.1 Firm Development in Knowledge Management

Source: M.A. White and G.D. Bruton, *The Management of Technology and Innovation*, 2nd Edition, Cengage Learning, 2011.



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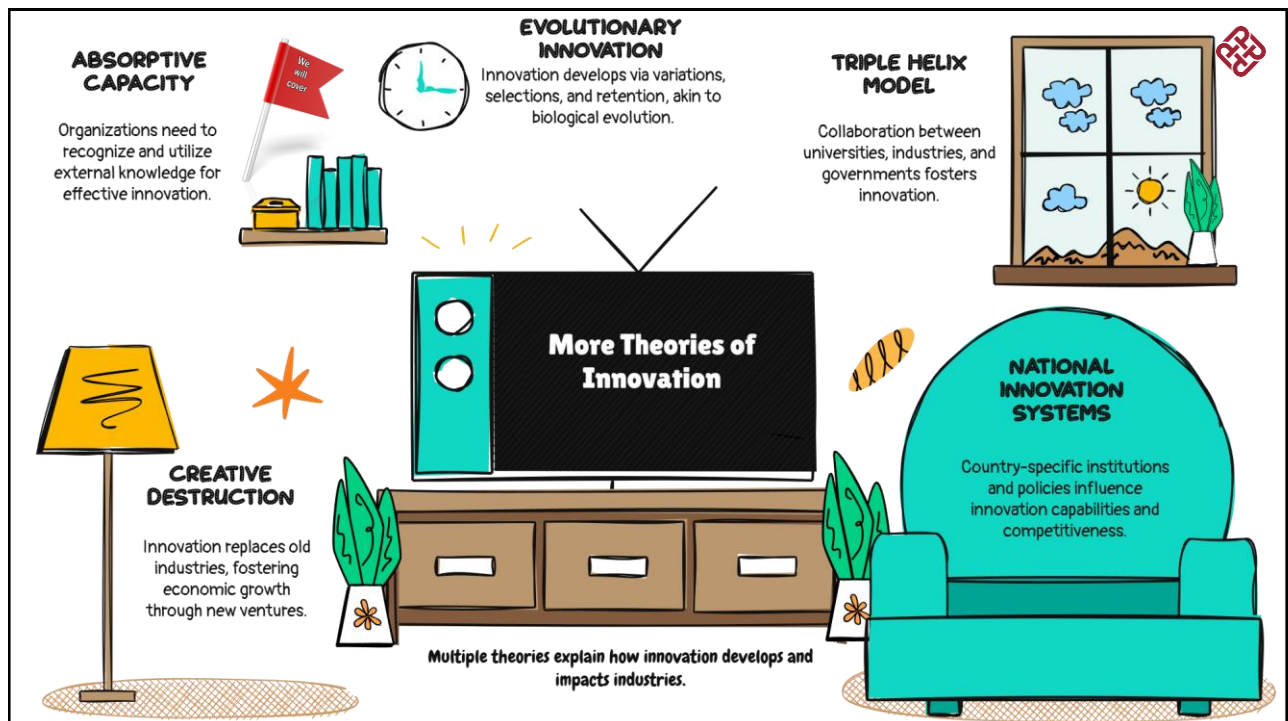
Successful innovators need to access a lot of information ...

- In-house research and development, including basic research
- Linkages to customers or other potential users of innovations
- Linkages to an external network of firms that may include competitors, complementors, and suppliers
- Linkages to other external sources of scientific and technical information, such as universities and government laboratories



Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

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Absorptive capacity

Absorptive capacity – a firm's ability to better assimilate and utilize information obtained externally. Absorptive capacity refers to the firm's ability to understand and use new information



The effects of absorptive capacity suggest that firms that develop new technologies ahead of others may have an advantage in staying ahead. Firms that forgo investment in technology development may find it very difficult or expensive to develop technology in a subsequent period.

Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

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Understanding Absorptive Capacity Theory

Cohen and Levinthal's theory explains innovation through learning and adaptation.

Definition of Absorptive Capacity

Absorptive capacity refers to an organization's ability to recognize, assimilate, and apply external knowledge effectively.

Three Key Components

The components include acquisition, assimilation, and exploitation of knowledge, forming a holistic approach to learning.

Importance for Innovation

It enhances firms' capacity to innovate, leading to sustained competitive advantage in dynamic environments.

Role of Prior Knowledge

A firm's existing knowledge base directly influences its ability to absorb new information efficiently.

Building Absorptive Capacity

Organizations can enhance their absorptive capacity through training, collaboration, and effective communication practices.

Impact of Organizational Culture

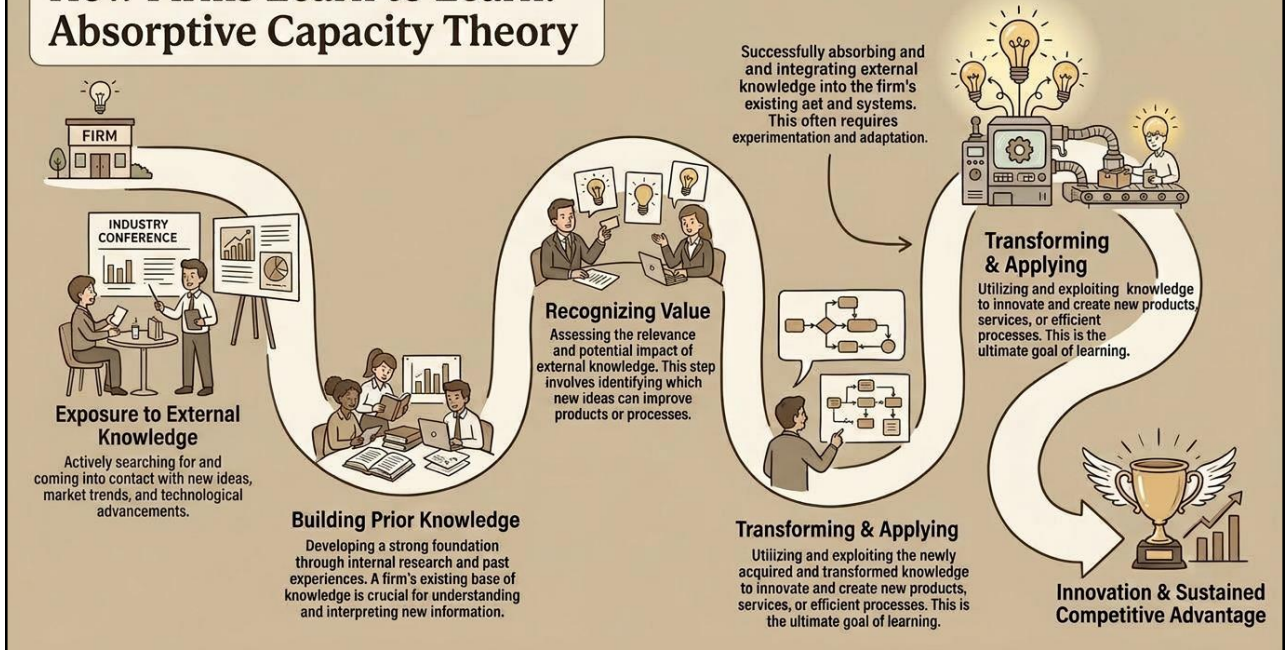
Cultivating a supportive culture promotes knowledge sharing, contributing to higher absorptive capacity and innovation success.



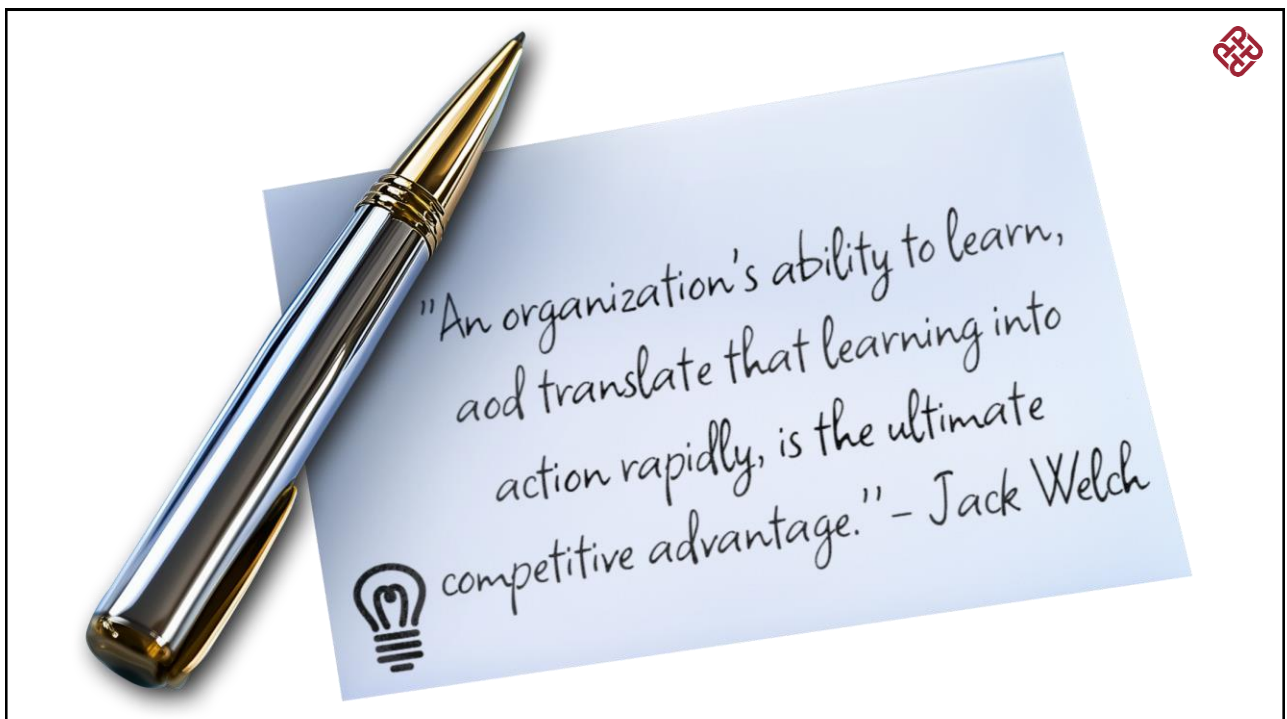
Source: Cohen, W. M., & Levinthal, D. A. (1990). *Absorptive Capacity: A New Perspective on Learning and Innovation*. Administrative Science Quarterly, 35(1), 128–152

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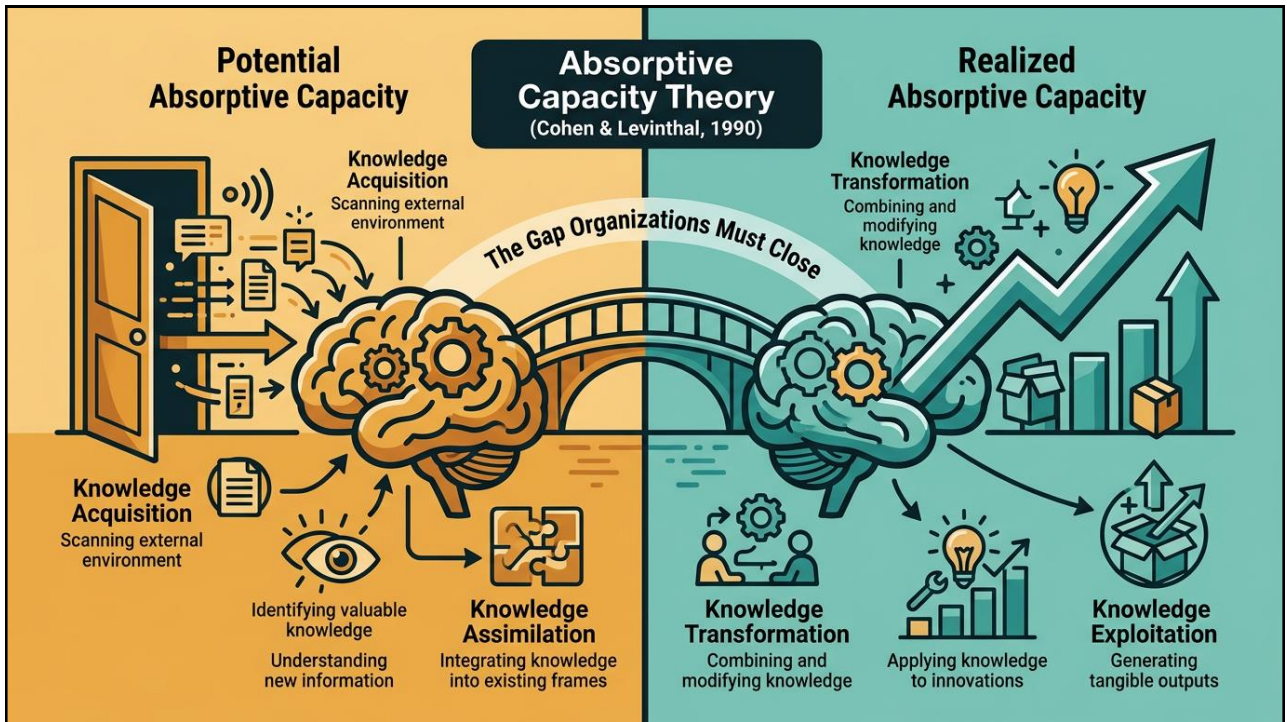
How Firms Learn to Learn: Absorptive Capacity Theory



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Learn from failures



It's through mistakes that you actually can grow. You have to get bad in order to get good." - Paula Scher



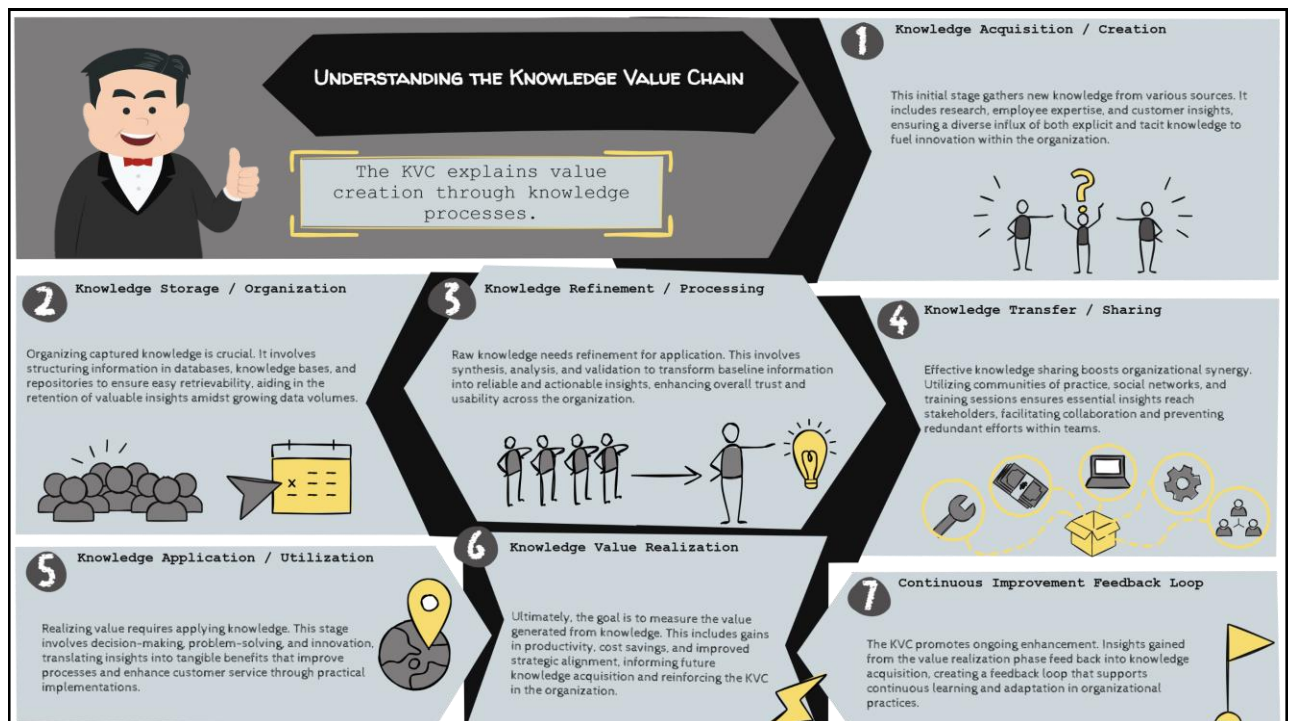
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Knowledge-based theory

Operationalising Knowledge Processes



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Linear and Non-linear systems



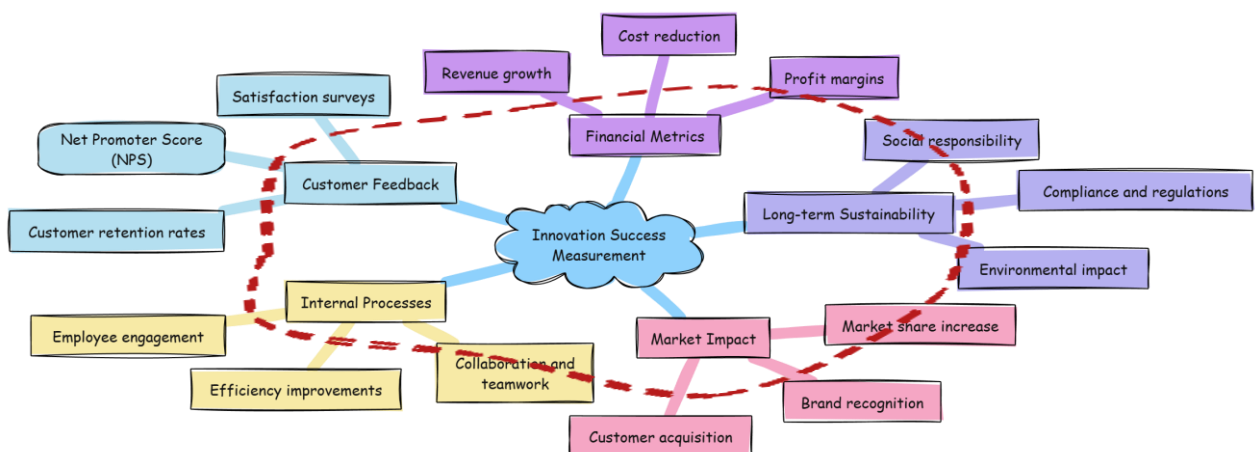
- A linear system is one where the output changes proportionally to the input.
- A non-linear system is one where the output does not necessarily change proportionally to the input.

Input	Linear System	Non-linear System
Double the input	Output doubles	Output may triple, quadruple, or change unpredictably
Example	10 workers produce 100 units, 20 workers produce 200 units	10 workers produce 100 units, 20 workers may produce only 170 units because they get in each other's way
Graph	Straight line	Curved line

An Innovation Ecosystem is a non-linear system

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Measuring the success of innovation



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What makes something a good idea?



Equation to use with valuable problems to solve

$$\text{Value of the potential opportunity} = \left(\text{Number of target customers} \times \text{Frequency that the solution is used} \times \text{Willingness to pay for current solution}^1 \right) \div \text{Level of satisfaction with current available alternatives}$$

¹Willingness to pay is a proxy for customer ascribing meaningful value. Whether companies choose to charge and who pays is a business-model decision.

(Source: Jordan Bar Am, Laura Furstenthal, Felicitas Jorge, and Erik Roth, *Innovation in a crisis: Why is it more critical than ever?*, McKinsey Quarterly, June 2020)

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